

A ‘ramp-sprint’ protocol to characterise indices of aerobic function and exercise intensity domains in a single laboratory test

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Abstract

Purpose The lactate threshold (LT), critical power (CP) and maximum oxygen uptake ($\dot{V}O_{2\max}$) together partition exercise intensity domains by their common physiological, biochemical and perceptual response characteristics. CP is the greatest power output attainable immediately following intolerance at $\dot{V}O_{2\text{peak}}$, and the asymptote of 3 min all-out exercise. Thus we reasoned that a maximal ‘sprint’ immediately following standard ramp-incremental exercise would allow characterisation of the three aerobic indices in a single test.

Methods Ten healthy men (23 ± 3 year, mean $\pm$ SD) performed 9 cycle-ergometry tests on different days: (A) two ramp-incremental tests to intolerance (20 W min^{-1}), immediately followed by a 3 min maximal, variable-power effort ramp-sprint test (RST) for LT, $\dot{V}O_{2\text{peak}}$ and sprint-phase power (SP) determination; (B) four constant-power tests for CP and $\dot{V}O_{2\max}$ determination; (C) constant-power tests at 10 W below LT, and 10 W below and above SP to verify

intensity domain characterisation. Capillary [lactate] and breath-by-breath $\dot{V}O_2$ were measured.

Results Reproducibility of LT, SP and $\dot{V}O_{2\max}$ measurements between RST repeats was within 5 % or less ($r \geq 0.991$, $p < 0.001$). CP ($257 \pm 46 \text{ W}$) was not different ($p = 0.72$) from SP ($258 \pm 42 \text{ W}$). Exercise 10 W below LT and SP resulted in steady state $\dot{V}O_2$ and [lactate]. $\dot{V}O_{2\max}$ ($4.0 \pm 0.6 \text{ L min}^{-1}$), peak [lactate] ($11 \pm 2 \text{ mM}$) and intolerance were reached 19 ± 5 min into exercise at 10 W above SP.

Conclusions These data suggest that the key indices of aerobic function may be accurately and reliably estimated during a single exercise test. This test may provide a basis for simplifying assessment and prescription of exercise training and experimental interventions.

Keywords Gas exchange · Maximum oxygen uptake · Exercise testing · Critical power · Lactate threshold

Abbreviations

ANOVA	Analysis of variance
CI ₉₅	95 % confidence interval
CP	Critical power
LoA	Limits of agreement
LT	Lactate threshold
P	Power output
P _{peak}	Peak power output
RST	Ramp-sprint test
SD	Standard deviation
SE	Standard error
SP	Sprint power
t _{lim}	Tolerable duration
VO ₂	Pulmonary oxygen uptake
$\dot{V}O_{2\max}$	Maximum oxygen uptake
$\dot{V}O_{2\text{peak}}$	Peak oxygen uptake

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W'	Curvature constant
WSP	Work done above SP
% Δ	Percentage difference between LT and P_{peak}

Introduction

The assignment of exercise intensity for individualised exercise assessment or prescription is best achieved by identifying the range of power outputs that produce common physiological strain in the muscle metabolic, pulmonary gas exchange and ventilation, and blood acid–base responses to constant-power exercise (e.g. Roston et al. 1987; Poole et al. 1988; Özyener et al. 2001; Jones et al. 2008). However, the most common normalisation procedures used, such as percentages of maximum heart rate, peak power production or maximum pulmonary oxygen uptake ($\dot{V}O_{2\text{max}}$), assessed during a standard incremental exercise test, are inadequate to produce uniform physiological responses among participants (Katch et al. 1978; Whipp et al. 2005; Lansley et al. 2011). This is because, among individuals, the parameters and variables that partition the range of potential power outputs into clusters with common physiological response characteristics—i.e. the lactate threshold (LT), critical power (CP) and $\dot{V}O_{2\text{max}}$ (Whipp 1996)—occur at widely-variable fractions of maximal values (e.g. Poole et al. 1988; van der Vaart et al. 2013). As such, assigning exercise intensity based on a given % $\dot{V}O_{2\text{max}}$, for example, may elicit markedly different responses among participants. This variance complicates the accurate application and interpretation of aerobic exercise assessment and prescription.

The ‘intensity domain’ schema proposed by Whipp (1996) (although see Hill et al. 2002 for an alternative classification) assigns a term to each exercise intensity (moderate, heavy, very-heavy and severe) that aligns closely with the sensory perceptions of effort (Scherr et al. 2012). Moderate intensity exercise denotes the power output (P) range for which steady states in oxygen uptake ($\dot{V}O_2$) are attained without a sustained metabolic acidosis, i.e. <LT (Whipp 1996; Özyener et al. 2001). In the heavy intensity domain, i.e. power outputs between LT and CP, attainment of a $\dot{V}O_2$ steady state is delayed by up to ~20 min. The corresponding increases in arterial [lactate] and $[H^+]$ eventually stabilise, but are accompanied by a reduced work efficiency relative to moderate exercise (e.g. Casaburi et al. 1987; Roston et al. 1987). In the very-heavy intensity domain (i.e. >CP), $\dot{V}O_2$, arterial [lactate] and $[H^+]$, muscle acidity, [inorganic phosphate] and phosphocreatine breakdown each continue progressively with $\dot{V}O_{2\text{max}}$ being attained at the tolerable limit (Poole et al. 1988; Jones et al. 2008). In the Whipp (1996) schema, power outputs for which the initial

whole-body ATP turnover rate exceeds that at $\dot{V}O_{2\text{max}}$ are classified as severe intensity (Özyener et al. 2001).

The current gold-standard for non-invasive cardiopulmonary exercise assessment is the ramp-incremental test (Whipp et al. 1981), which provides measurements of LT and $\dot{V}O_{2\text{peak}}$ thus demarcating the moderate and severe intensity domains (Whipp 1996). The ramp-incremental test also provides additional key indices of aerobic function, such as functional gain (the ratio of the change in $\dot{V}O_2$ to power output; $\Delta\dot{V}O_2/\Delta P$) and the $\dot{V}O_2$ mean response time (Whipp et al. 2005), as well as important integrated diagnostic and prognostic cardiopulmonary variables such as the oxygen pulse ($\dot{V}O_2/\text{HR}$) and the ventilatory equivalent for CO_2 ($\dot{V}_E/\dot{V}CO_2$) (Palange et al. 2007; Wasserman et al. 2012). This test, however, does not provide measurement of CP, and so is unable to fully characterise the domains of aerobic function in a single laboratory visit.

The relationship between power output and the tolerable exercise duration (t_{lim}) has been well described as hyperbolic for many exercise modalities, including cycle ergometry (Poole et al. 1988; Hill et al. 1995; Morton et al. 1997; Dekerle et al. 2003; Coats et al. 2003; Burnley et al. 2006; Jones et al. 2010; van der Vaart et al. 2013). CP is the asymptote of this relationship, with a metabolic equivalent that occurs halfway between LT and $\dot{V}O_{2\text{max}}$ on average, but which varies widely among participants (van der Vaart et al. 2013). The curvature constant of this relationship (W') represents a finite amount of supra-CP work that can be tolerated during constant-power (Fukuba et al. 2003), variable-power (Vanhatalo et al. 2007) and ramp-incremental (Morton et al. 1997) exercise. Therefore, the highest power output sustainable for more than a few seconds immediately after intolerance at $\dot{V}O_{2\text{peak}}$ in a ramp-incremental exercise, i.e. following depletion of W' , should be equal to CP (Morton et al. 1997; Coats et al. 2003; Burnley et al. 2006; Brickley et al. 2007).

We therefore aimed to determine whether exercise intensity domains could be estimated from a single exercise test, by measuring the power output during a maximal effort (i.e. a ‘sprint’), performed immediately after the tolerable limit of standard ramp-incremental exercise on a cycle ergometer. Specifically, the initial ramp-incremental phase would provide estimates of LT and $\dot{V}O_{2\text{max}}$ (Whipp et al. 1986) while serving to deplete W' (Morton et al. 1997), with the subsequent sprint phase generating a mean sprint power (SP) notionally equivalent to CP (Vanhatalo et al. 2007). Given that the ramp-incremental phase was expected to deplete W' , we reasoned that the necessary sprint duration to establish a stable SP measurement would be short. However, the duration necessary to achieve this stability of SP is unknown. Based on the findings of Burnley et al. (2006)

and Vanhatalo et al. (2007), we selected a 3 min sprint duration as being sufficient for attainment of CP.

To investigate the efficacy of this ‘ramp-sprint test’ (RST) protocol to identify the domains of aerobic function in a single laboratory test, we evaluated the following hypotheses: (1) the single-test RST would provide reproducible estimates of LT, SP and $\dot{V}O_{2\text{peak}}$; (2) the SP and $\dot{V}O_{2\text{peak}}$ resulting from the single-test RST would be equal to CP and $\dot{V}O_{2\text{max}}$ resulting from conventional testing procedures; and (3) constant power exercise tests at power outputs derived from the single-test RST would result in physiological responses characteristic of exercise within moderate, heavy and very-heavy intensity domains. In testing these hypotheses we also aimed to identify the duration required for SP to stabilise during the sprint phase of the test.

Methods

Participants

Ten healthy, habitually-active men [mean $\pm$ standard deviation (SD); age: 23 ± 3 year; height: 179 ± 8 cm; body mass: 75 ± 11 kg] took part in the investigation after providing written informed consent, as approved by the University of Leeds Ethical Review Committee. Of the ten, seven had never previously performed a laboratory exercise test and none were engaged in formal training at the time of the study. Following a familiarisation ramp-incremental test (Whipp et al. 1981), participants visited the air-conditioned laboratory (20 ± 1 °C) on at least nine occasions, each at a similar time of day (± 1 h) on non-consecutive days; all procedures were in accordance with the *Declaration of Helsinki*. Participants were instructed to be well hydrated and rested (no strenuous exercise in the previous 24 h) and to refrain from alcohol consumption (24 h) and food and caffeine ingestion (3 h) before each testing session.

Equipment and measurements

All exercise tests were conducted using an electromagnetically-braked cycle ergometer that featured both cadence-dependent and cadence-independent power output control (Excalibur Sport; Lode, Groningen, NL). External power output and pedal cadence from the ergometer were digitised by a PowerLab 8/30 data acquisition system (ADInstruments, Oxford, UK) and recorded at 2 Hz using LabChart 5 Pro. Heart rate from the R-R interval of a 12-lead ECG (Burdick Quest; Cardiac Science, Sale, UK) and arterial oxygen saturation via earlobe pulse oximetry (Biox 3745; Ohmeda, Louisville, KY) were also recorded on this system.

Inspired and expired flows and volumes were measured using a low dead space (90 mL), low resistance (<0.65 cmH₂O at 8.5 L s⁻¹) turbine volume transducer (Interface Associates, Laguna Niguel, CA) connected to a mouth-piece. Respired gas, sampled continuously from the mouth-piece at 1 mL s⁻¹, was analysed for O₂, CO₂ and N₂ concentrations by mass-spectrometry (MSX; nSpire Health, Hertford, UK). The turbine and mass-spectrometer were calibrated directly before each test using known volumes of air over a range of flows and precision-analysed gas mixtures spanning the range of the inspired and expired gas concentrations, respectively. The time delay between the volume and gas concentration signals was resolved automatically by the manufacturer’s algorithms. The analogue signals were digitised every 20 ms and time-aligned for breath-by-breath calculation of pulmonary gas exchange and ventilatory variables using the algorithms of Beaver et al. (1973). The precision-analysed gas mixtures were re-sampled immediately after each test to verify the calibration stability.

During selected protocols, serial ~ 25 μ L samples of arterialed capillary blood were taken from the fingertip of a heated hand for the analysis of [lactate] using an automated analyser (GM7; Analox Instruments, London, UK). The analyser was calibrated with an 8 mM standard before, and verified at regular intervals throughout analysis.

Exercise protocols

All tests were preceded (≥ 4 min) and followed (≥ 6 min) by cycling at 20 W. During ramp-incremental exercise participants steadily increased their pedal cadence to a self-selected rate (range 75–95 rpm) and thereafter maintained it within ± 5 rpm. If at any point during a test the cadence began to fall from the lower bound of this range, participants were given strong verbal encouragement to recover their cadence back to their self-selected rate. Exercise intolerance was defined as the point at which participants were unable to maintain a cadence of at least 50 rpm.

Ramp-sprint test (RST). Each participant performed two RSTs. During the initial ramp phase power increased as a smooth function of time at 20 W min⁻¹, and was continued to the limit of tolerance for estimation of LT, $\dot{V}O_{2\text{peak}}$ and peak power output (P_{peak}). Immediately following ramp intolerance, the cycle ergometer was switched to its cadence-dependent (linear) mode and participants were given strong verbal encouragement to quickly accelerate their cadence and deliver a maximal ‘sprint’ effort. This sprint phase was continued for 3 min to estimate the sprint power (SP); 3 min was chosen on the basis that power output has been previously demonstrated for cycle ergometry to consistently stabilise within this duration (Burnley et al. 2006; Vanhatalo et al. 2007), thus allowing post hoc

analysis of the minimum duration required to stabilise SP. Resistance to pedalling was standardised across participants using the linear mode of the ergometer, where the programmed 'linear factor' determines the flywheel braking resistance from power output/cadence². As CP is a metabolic rate that includes both internal and external power production (Barker et al. 2006), its measurement by external power alone will be sensitive to variations in internal (unmeasured) power production that could occur when cadence is allowed to vary excessively (Hill et al. 1995; Barker et al. 2006; Vanhatalo et al. 2008b; Dekerle et al. 2013). In the RST, however, it is necessary to set the flywheel resistance prior to the exercise test. For this we first determined the relationship between CP and body mass derived from 60 healthy young participants (van der Vaart et al. 2013). From this relationship, CP (measured in W) was approximated by three times body mass (in kg). The power output value resulting from this approximation in each individual (e.g. 240 W in an 80 kg participant) was used in the equation to set the flywheel resistance at a cadence of 80 rpm (also see Bergstrom et al. 2012; Clark et al. 2013).

Constant-power tests. Four constant-power tests, each at different work rates, were completed to the limit of tolerance. Power outputs were selected to elicit intolerance within the range of ~3–15 min allowing estimation of CP, W' and $\dot{V}O_{2\max}$ using standard methods (Poole et al. 1988; Day et al. 2003). CP and W' were established from the intercept and slope, respectively, of the linear power – tolerable duration⁻¹ ($P - t_{\lim}^{-1}$) relationship ($P = [W'/t_{\lim}] + CP$) using least-squares linear regression. An additional (5th) test was performed in three participants to reduce the standard error (SE) of CP and W' estimates.

Participants performed three further constant-power tests for a 30 min target duration or to intolerance (whichever came first). These were completed at power outputs equivalent to 10 W below LT, 10 W below SP and 10 W above SP, i.e. predicted by the RST to be moderate, heavy and very-heavy intensity, respectively. Arterialised-capillary blood was sampled at rest, during the 20 W baseline, every 5 min throughout exercise and (where applicable) immediately following intolerance. A steady state was identified for each participant where there was no further appreciable change in $\dot{V}O_2$ (≤ 0.2 L min⁻¹; Lamarra et al. 1987; Day et al. 2003) and [lactate] (≤ 1 mM; Heck et al. 1985) between successive measurements, with these being maintained until end-exercise at 30 min. The constant-power tests were completed in a counterbalanced order.

Data analyses

For each test, breath-by-breath pulmonary gas exchange data were edited to exclude infrequent aberrant breaths

(as triggered by swallows, coughs, sighs, etc.) which were ≥ 4 SD from the local mean. These data were considered uncharacteristic of the underlying physiological response (Lamarra et al. 1987). LT was estimated non-invasively from pulmonary gas exchange and ventilatory profiles in the RST using the V-slope method (Beaver et al. 1986), with supporting evidence from the dynamics of O₂ and CO₂ end-tidal gas fractions, ventilatory equivalents for O₂ and CO₂ and the respiratory exchange ratio (Whipp et al. 1986). For exercise tests performed to the limit of tolerance, $\dot{V}O_{2\text{peak}}$ was calculated as the mean $\dot{V}O_2$ for an integral number of breaths over the final 20 s, and $\dot{V}O_{2\max}$ was subsequently established by comparison across tests that utilised different power outputs (Day et al. 2003).

$\dot{V}O_2$ and power output responses during the 3 min sprint phase of the RST were averaged into 30 s time-bins for time-dependent comparisons e.g. 1–30, 31–60 s and so on. This also enabled direct comparisons with the $\dot{V}O_{2\max}$ and CP, respectively. The time-integrated power output above SP throughout the RST, i.e. during both the ramp and sprint phases, was calculated to provide an estimate of W' .

Statistics

Parameters and variables from the two RSTs (LT, SP, $\dot{V}O_{2\text{peak}}$ and P_{peak}) were compared using paired samples t tests, within-participant coefficients of variation and intra-class correlation coefficients. One-way repeated measures ANOVA was performed to identify within-test differences in the 30 s time-bins of $\dot{V}O_2$ and power output during the 3 min sprint phase. Where a significant difference was observed, i.e. $p < 0.05$, Tukey's post hoc test was used to identify the specific effects. Agreement between CP and SP was assessed by Bland–Altman mean bias ± 95 % limits of agreement (LoA) analysis. Data are reported as mean $\pm$ SD, unless otherwise stated.

Results

Ramp-sprint test (RST)

An example of the power output, $\dot{V}O_2$ and pedal cadence responses during the RST from an individual participant are shown in Fig. 1. Between the repeated RST tests there were no differences in estimates of LT (1.98 ± 0.28 vs. 1.96 ± 0.29 L min⁻¹, $p = 0.49$), or measurements of SP (257 ± 43 vs. 259 ± 43 W, $p = 0.15$), $\dot{V}O_{2\text{peak}}$ (4.03 ± 0.65 vs. 4.01 ± 0.59 L min⁻¹, $p = 0.55$) and P_{peak} (344 ± 52 vs. 342 ± 50 W, $p = 0.38$). These measurements also demonstrated high intra-participant precision: the test–retest difference for LT, SP, $\dot{V}O_{2\text{peak}}$ and P_{peak} was less than or equal to 5, 3, 4 and 5 %, respectively (Fig. 2), with low

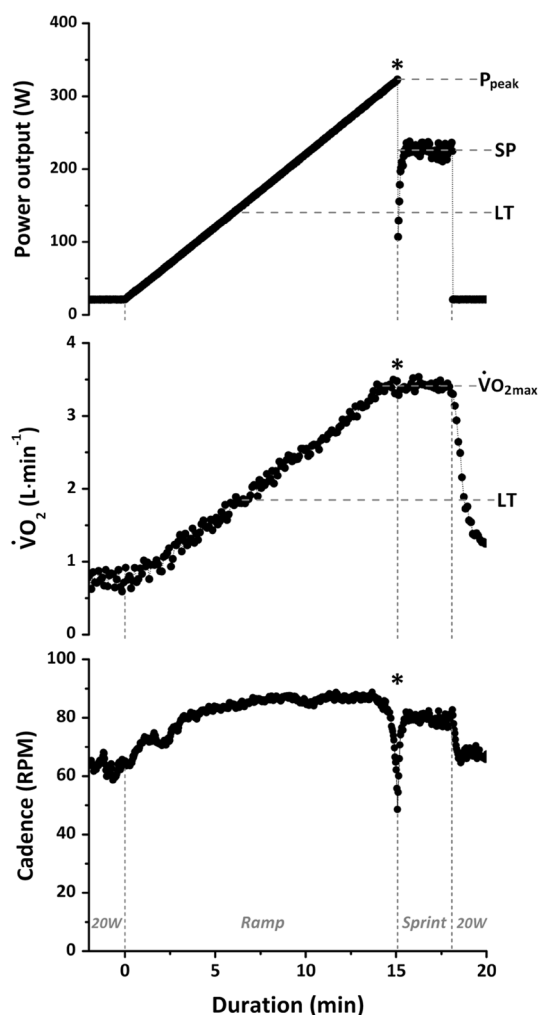


Fig. 1 The power output (*top*), $\dot{V}O_2$ (*middle*) and pedal cadence (*bottom*) responses of a representative participant to the ramp-sprint test (RST). Following baseline cycling at 20 W, power output was ramped as a smooth function of time at 20 W min⁻¹ and exercise continued to the limit of tolerance (intolerance is indicated by *asterisk*). Immediately following this, the participant completed a 3 min maximal ‘sprint’, during which the braking force on the flywheel was held constant and the power output was dependent on cadence. Power output was then returned to a 20 W baseline for recovery. P_{peak} peak ramp power, SP sprint power, LT estimated lactate threshold, $\dot{V}O_{2max}$ maximum oxygen uptake

coefficients of variation ($\leq 5\%$) and significant intra-class correlation coefficients ($r \geq 0.991$, $p < 0.001$) between all measurements for the two tests. Consequently, the mean RST values from each participant are reported in Table 1.

Following intolerance in the ramp-incremental phase of the RST, power output was rapidly recovered by switching to the ergometer’s cadence-dependent mode. Power output varied during the 3 min sprint effort ($F = 9.80$, $p = 0.001$, $\eta^2 = 0.52$), being significantly lower ($p < 0.001$) in the first 30 s compared to all other times (Fig. 3a). There was no discernible change in the mean power output in the time-bins

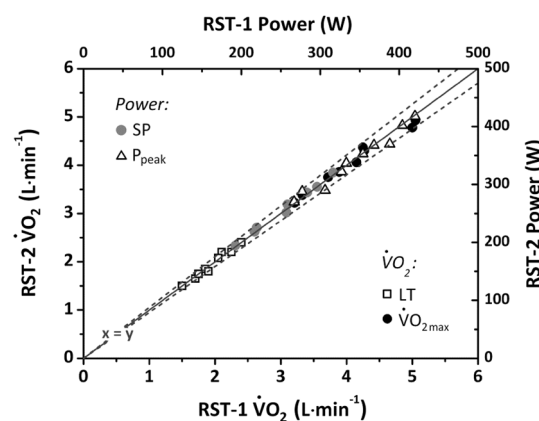


Fig. 2 A cross-plot illustrating the reproducibility of parameter and variable estimates between the repeated ramp-sprint tests (RST-1 vs. RST-2). LT and $\dot{V}O_{2max}$ (L min⁻¹, *bottom and left axes*) and SP and P_{peak} (W, *top and right axes*) were highly reproducible between tests. Dotted lines show 5% difference limits

between 30 and 180 s of the RST sprint phase, and therefore SP was calculated from the mean power output over this period. SP averaged 258 ± 42 W (Table 1) with a 95% confidence interval (CI₉₅) of 2 ± 2 W. Although SP averaged only $75 \pm 4\%$ of P_{peak} , $\dot{V}O_2$ was maintained at the ramp phase peak throughout the RST sprint phase. There were no differences ($F = 0.36$, $p = 0.87$, $\eta^2 = 0.04$) in $\dot{V}O_{2peak}$ values between the ramp and sprint phase (Table 1) or among 30 s $\dot{V}O_2$ time-bins across the sprint phase (Fig. 3b). SP was achieved at a mean cadence of 81 ± 3 rpm (Table 1; Fig. 3c). The power output – time integral above SP during the RST was 13.3 ± 4.1 kJ.

It should be noted that one participant demonstrated an early overshoot in power output during the initial ~90 s of the sprint phase, unlike the other nine participants whose responses closely resembled Figs. 1 and 3 (no differences were apparent in this participant’s physiological characteristics, however). The response for this individual was the same for RST-1 and RST-2, and $\dot{V}O_{2peak}$ was maintained throughout the sprint phase, suggestive of a continued maximal effort. In this participant, SP was therefore estimated from the mean power output over the final 60 s of the sprint, i.e. where power had clearly levelled-out.

Constant-power tests

Individual $\dot{V}O_{2peak}$ values (mean 4.04 ± 0.60 L min⁻¹) did not vary among constant-power tests to intolerance, nor did they differ from those achieved during the RST ($F = 0.40$, $p = 0.85$, $\eta^2 = 0.04$). This confirmed the $\dot{V}O_{2peak}$ values during the RST as representing $\dot{V}O_{2max}$ (Table 1) (Hill and Lupton 1923; Day et al. 2003). The $P - t_{lim}^{-1}$ relationship was well modelled by a linear function in all cases ($r^2 > 0.98$) (e.g. Fig. 4a). CP and W' from this model

Table 1 Individual responses, parameters and variables of aerobic function from the RST

Participant	Ramp phase					Sprint phase				
	LT		$\dot{V}O_{2\text{peak}}$		P_{peak}	SP		Cadence	$\dot{V}O_{2\text{max}}$	
	L min ⁻¹	W	L min ⁻¹	mL kg ⁻¹ min ⁻¹		W	%Δ	rpm	L min ⁻¹	
1	2.23	158	4.31	59	403	296	57	85	4.44	
2	2.40	165	4.99	68	419	318	60	84	4.86	
3	2.23	165	4.89	57	379	301	64	86	4.94	
4	1.85	140	3.41	54	324	226	46	80	3.41	
5	2.15	145	4.10	51	368	262	52	78	3.92	
6	1.68	115	3.35	51	283	223	64	79	3.45	
7	1.50	95	3.22	50	268	194	57	77	3.12	
8	1.85	115	3.73	57	298	218	56	85	3.68	
9	1.75	125	3.87	46	335	254	62	79	3.88	
10	2.08	150	4.30	44	353	286	67	80	4.22	
Mean	1.97	137	4.02	54	343	258	59	81	3.99	
SD	0.29	24	0.62	7	51	42	6	3	0.62	

RST ramp-sprint test, LT estimated lactate threshold, $\dot{V}O_{2\text{peak/max}}$ peak/maximum oxygen uptake, P_{peak} peak ramp power, SP sprint power, %Δ percentage difference between LT and P_{peak} , Cadence pedal cadence at which SP was achieved

averaged 257 ± 46 W and 17.0 ± 4.1 kJ, respectively, with high confidence in the parameter estimates verified by low SE (CP: 2.0 ± 0.8 W, W' : 0.6 ± 0.2 kJ) and CI_{95} values (CP: 7.5 ± 2.0 W, W' : 2.2 ± 0.7 kJ). There was no difference between CP and SP estimates ($p = 0.72$, mean bias 0.9 W, $CI_{\text{Difference}}$ -4.7 , 6.5 W), with narrow limits of agreement within-participants: 95 % LoA -14 , 16 W (Fig. 4b), equivalent to -5 , 6 % of CP. The work done above SP in the RST was less than W' ($p = 0.02$, mean bias -3.7 kJ, $CI_{\text{Difference}}$ -6.5 , -0.8 kJ) (Fig. 4c).

Figure 5 shows the group mean responses to constant-power tests at 10 W below LT, 10 W below SP and 10 W above SP. All ten participants comfortably completed 30 min of exercise at 10 W below LT, with steady states being achieved in $\dot{V}O_2$ (1.74 ± 0.30 L min⁻¹ attained after 5 min) and [lactate] (1.3 ± 0.4 mM after 5 min) (Fig. 5). All participants also completed 30 min at 10 W below SP (which equated to 73 ± 4 % of P_{peak}), with delayed steady states in $\dot{V}O_2$ (attained after 5 min in $n = 1$, 10 min in $n = 8$, 20 min in $n = 1$; 3.52 ± 0.30 L min⁻¹) and [lactate] (after 10 min in $n = 5$, 15 min in $n = 2$, 20 min in $n = 3$; 8.1 ± 1.2 mM). Only one participant reached the 30 min target during exercise at 10 W above SP, and this was achieved in the absence of a steady state. On average, exercise at 10 W above SP resulted in the attainment of $\dot{V}O_{2\text{max}}$ (4.04 ± 0.58 L min⁻¹) and a peak [lactate] of 10.7 ± 1.5 mM after 19 ± 5 min.

Discussion

This study examined the efficacy of a 'ramp-sprint test' (RST) to characterise exercise intensity domains in a single laboratory test. In a group of healthy active young men the

cycle ergometer RST protocol provided estimates of LT, CP and $\dot{V}O_{2\text{max}}$ that were each highly reproducible (hypothesis 1) and accurate (hypothesis 2) to within ≤ 5 % compared to gold-standard procedures (Whipp et al. 1986; Poole et al. 1988; Day et al. 2003). The validity of the intensity domain boundary estimates from the RST were confirmed in the characteristic $\dot{V}O_2$ and [lactate] response profiles for constant-power cycling tests (hypothesis 3). Put together, these data provide what we believe to be the first demonstration that the demarcation of the moderate, heavy, very-heavy and severe intensity domains (Whipp 1996) can be reliably estimated during a single laboratory exercise test.

Characterising exercise intensity domains

There is currently no universally-agreed method for the assignment or normalisation of exercise intensity. Absolute and relative measures of, for example, heart rate, power output, metabolic equivalents (METs) and perceived exertion (RPE) have the advantage of ease of use (ACSM 2009). However, they lack rigour for accurate normalisation of exercise intensity among participants differing in age, sex, state of training or health. Even the most widely used measure, % $\dot{V}O_{2\text{max}}$, has been criticised because it is unable to account for the wide variation in LT (ranging 40–85 % $\dot{V}O_{2\text{max}}$) and CP (ranging 50–95 % $\dot{V}O_{2\text{max}}$) (e.g. Moritani et al. 1981; Poole et al. 1988; Neder et al. 2000; Wasserman et al. 2012; Barker et al. 2006, 2012; van der Vaart et al. 2013) among individuals, around which physiological, biochemical and perceptual responses to exercise are known to vary (Poole et al. 1988; Whipp 1996; Jones and Killian 2000). In other words, exercise intensity is not a linear function of % $\dot{V}O_{2\text{max}}$, or other commonly used variables. Whipp (1996) therefore proposed a scheme that

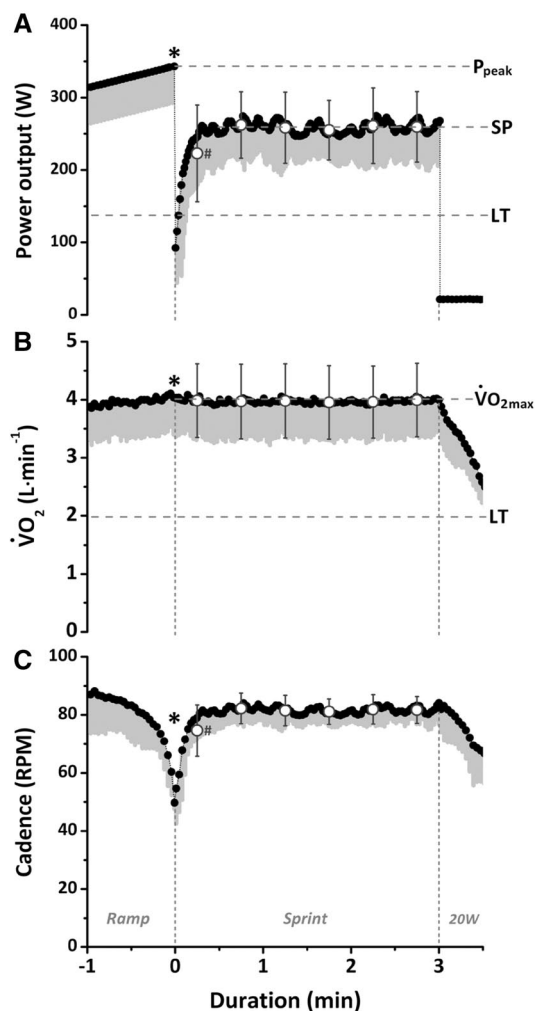


Fig. 3 Group mean (filled circles, with SD in grey) responses of power output (a), $\dot{V}O_2$ (b) and pedal cadence (c) during the ramp-sprint test (RST), plotted relative to the start of the 3 min sprint phase. Sprint responses were averaged into 30 s time-bins (unfilled circles, with SD bars) to facilitate sequential comparisons. Following ramp intolerance (intolerance is indicated by asterisk) the cycle ergometer was switched instantaneously from its cadence-dependent to its cadence-independent mode. Power output and pedal cadence were rapidly recovered and then sustained relatively constant for the remainder of the 3 min sprint. Despite SP only attaining 75 ± 4 % of P_{peak} , $\dot{V}O_2$ was maintained at its maximum throughout the sprint phase. # significantly different ($p < 0.05$) from the other time-bins

assigns common features of cardiopulmonary, neuromuscular and acid–base exercise responses into one of four intensity domains. However, there existed no assessment that could identify all three indices (LT, CP and $\dot{V}O_{2max}$) demarcating these domains in a single test.

The ramp-incremental test is the gold-standard for exercise testing as it provides a wide range of diagnostic and prognostic information in a single test, including LT and $\dot{V}O_{2peak}$, whilst minimising the duration spent approaching, or at, $\dot{V}O_{2max}$ (Whipp et al. 1981; Whipp and Ward 1992;

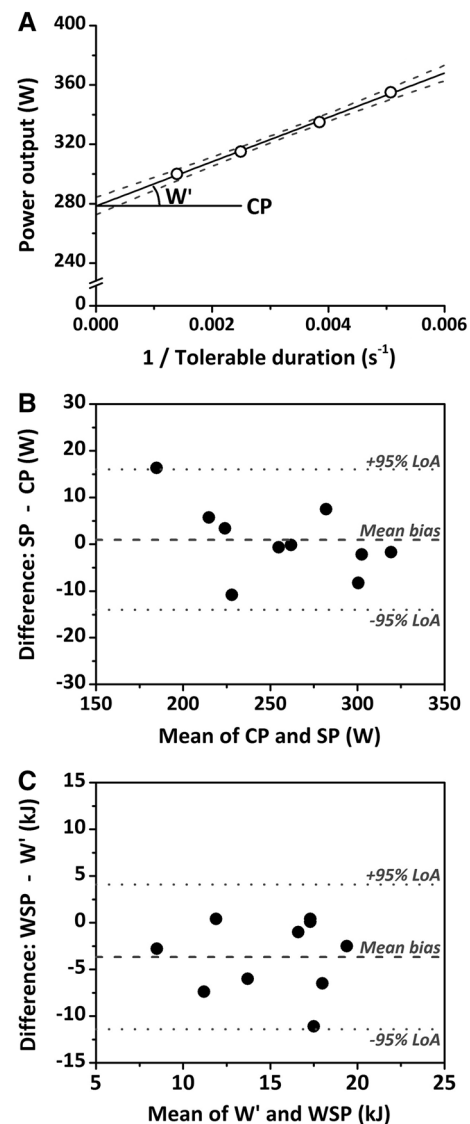


Fig. 4 The power – duration⁻¹ ($P - t_{lim}^{-1}$) relationship (a) for an individual participant as described by four different constant-power tests (unfilled circles), each taken to the limit of tolerance. Linear regression (solid line) with 95 % confidence limits (CI₉₅; dashed lines) was used to determine CP (the power-intercept) and W' (the slope) of the relationship. A Bland–Altman analysis showing the agreement between CP and SP (b) for each participant (filled circles), the mean bias (dashed line) ± 95 % limits of agreement (LoA, dotted lines) are illustrated. A Bland–Altman analysis showing the agreement between W' and work done above SP (c) for each participant (filled circles), the mean bias (dashed line) ± 95 % limits of agreement (LoA, dotted lines) are illustrated. CP critical power, W' curvature constant, SP sprint power, WSP work done above SP

ATS and ACCP 2003; Palange et al. 2007). The ramp-incremental test, however, does not provide a measure of CP and is therefore unable to fully account for variations in intensity domains among individuals. Attempts to overcome this limitation have led to the use of the ‘percentage delta’ power output range ($\% \Delta$; Casaburi et al. 1987;

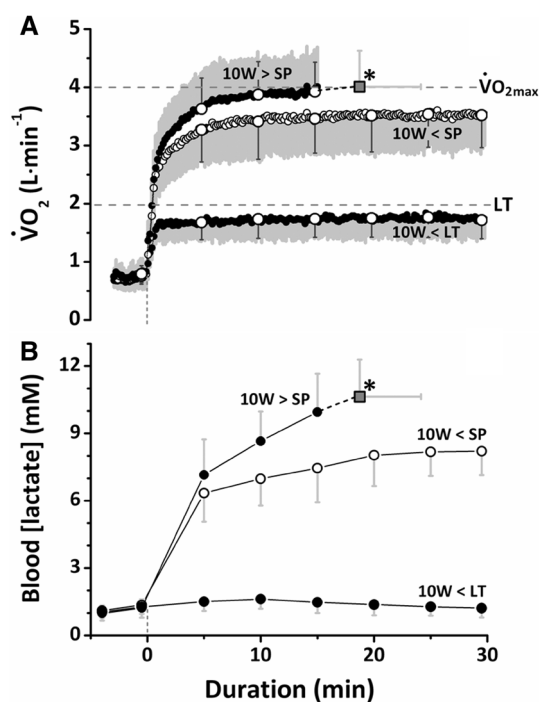


Fig. 5 Group mean $\pm$ SD responses of $\dot{V}O_2$ (a) and blood [lactate] (b) to constant-power exercise at 10 W below LT, 10 W below SP and 10 W above SP. Both $\dot{V}O_2$ and [lactate] measurements attained a steady state at 10 W below LT and SP, with all participants reaching the 30 min target. Exercise at 10 W above SP was tolerated for an average of 19 ± 5 min, with $\dot{V}O_2$ and blood [lactate] rising throughout the test until the point of intolerance in all cases (intolerance is indicated by asterisk)

Roston et al. 1987; Lansley et al. 2011), i.e. describing the power outputs between LT and P_{peak} as 0–100 % Δ . Exercise prescription based on % Δ can reduce the variability among subjects in pulmonary gas exchange, ventilation and capillary [lactate] responses compared to methods based on % $\dot{V}O_{2max}$ (Casaburi et al. 1987; Roston et al. 1987; Rausch et al. 1991; Lansley et al. 2011). Even so, while CP averages ~ 50 % Δ for groups, it can range from 10 to 70 % Δ across individuals (Poole et al. 1988; Ferguson et al. 2007; Murgatroyd et al. 2011; Barker et al. 2012; van der Vaart et al. 2013; present study), meaning that intensity normalisation or exercise prescription using % Δ , while an improvement on traditional methods, remains open to imprecision.

The addition of CP measurement into a single-test RST therefore provides a potential solution to simplify exercise assessment, to improve accuracy of exercise prescription and for normalising exercise intensity for research studies. Alternative approaches to this problem have proved either unsuccessful (Carter et al. 1999), or have been based on the incorrect notion that the onset of ventilatory compensation in the ramp-incremental test provides an estimate of CP (Dekerle et al. 2003; Pouilly et al. 2005). That is, rather

than being a metabolic construct, the existence of the ventilatory compensation point for ramp-incremental exercise is nothing more than a reflection of the slow dynamics of the compensation, such that it is actually coincident with the LT for slow work-rate incrementation rates but variably delayed with respect to the LT for rapid incrementation rates depending on peripheral chemoreflex responsiveness (Wasserman et al. 2012). Recently, Constantini et al. (2014) demonstrated that estimates of the parameters demarcating the exercise intensity domains may be characterised during a single study visit using two separate tests separated by 20 min of recovery. Our data advance this notion to provide strong support for the efficacy of the RST protocol to assess exercise intensity domains in a single ~ 15 min exercise test.

The ramp-sprint test

The basis for the RST is well founded. Morton et al. (1997) showed that CP and W' were not different, whether derived from serial ramp-incremental or constant-power tests, suggesting that ramp-incremental exercise is sufficient to deplete W' . In addition, the power-duration model suggests that CP should be the greatest achievable power following W' depletion (Moritani et al. 1981; Chidnok et al. 2013). Consistent with this notion we and others have demonstrated, to within 10 %, that the greatest power output tolerated for more than a few seconds following W' depletion at $\dot{V}O_{2max}$ is not different from CP (Coats et al. 2003; Burnley et al. 2006; Vanhatalo et al. 2007). The challenge then becomes how best to measure the greatest power output achievable following the limit of tolerance in standard ramp-incremental exercise.

For this we adopted the novel approach developed by Burnley et al. (2006) of a sustained (3 min) all-out exercise test, for which the instantaneous power output is shaped by the participant through the product of a constant flywheel resistance and a variable self-determined cadence. Three minutes of all-out exercise was found to be sufficient in healthy young participants to produce an asymptotic power output at the end of the test that did not differ from CP, the work done above which was not different from W' (Vanhatalo et al. 2007). This approach was subsequently shown to estimate CP for a range of exercise modalities including cycling, treadmill running, cross country skiing and rowing (Francis et al. 2010; Johnson et al. 2011; Cheng et al. 2012; Broxterman et al. 2013; Fukuda et al. 2014), and also to be able to detect training-induced increases in CP (Vanhatalo et al. 2008a).

However, there has been some debate whether the 3 min all-out test performed in isolation provides valid estimates of CP (McClave et al. 2011; Barker et al. 2012; Dekerle et al. 2013; Karsten et al. 2013; Bergstrom et al. 2013,

2014) and of W' , for which the estimate from the all-out test is typically lower than that from the traditional series of constant-power tests (Cheng et al. 2012; Broxterman et al. 2013; Dekerle et al. 2013; Fukuda et al. 2014). Nevertheless, the CP and W' parameter estimates derived from the 3 min all-out test have been demonstrated to be sufficiently robust to predict laboratory or field cycling performance (Black et al. 2013; Chidnok et al. 2013). Additionally, CP estimates from the all-out test were not different when preceded by a ramp-incremental test and 20 min recovery (Constantini et al. 2014). As such, we reasoned that the combination of a ramp-incremental test and a 3 min all-out sprint in immediate series (by switching the ergometer from an experimenter-controlled hyperbolic mode at the end of the ramp to a participant-dependent linear mode), should allow LT, CP and $\dot{V}O_{2\max}$ to be assessed in a single laboratory test.

Our data substantiate this suggestion: RST-derived estimates of LT, CP and $\dot{V}O_{2\max}$ were highly reproducible (Fig. 2) and accurate compared with gold-standard measurements (Table 1; Figs. 4, 5). The 'sprint power' in the RST (SP; 258 ± 42 W) provided a close approximation of the independently determined CP (257 ± 46 W) in all ten participants (Fig. 4). The confidence limits for both CP (from $P - t_{\lim}^{-1}$) and SP measurement (from variance around the mean during the sprint phase) were within 5 % (10 W) in all participants, reflecting the best accuracy that the traditional gold-standard technique can deliver (Poole et al. 1988; Smith and Hill 1993; Ferguson et al. 2007; Murgatroyd et al. 2011). This accuracy was confirmed in separate constant-power tests at 10 W below and above SP (Fig. 5). All participants attained a delayed steady state in $\dot{V}O_2$ and [lactate] during exercise at 10 W below SP, consistent with sub-CP exercise in the heavy intensity domain (Poole et al. 1988). None of the 10 participants, however, were able to attain $\dot{V}O_2$ and [lactate] steady states during exercise at 10 W above SP, and 9 of the 10 participants reached intolerance at $\dot{V}O_{2\max}$ before the 30 min target; responses consistent with supra-CP exercise in the very-heavy intensity domain (Whipp 1996). This validation of SP, by two independent means, therefore provides strong evidence that the single-visit RST allows valid and accurate CP estimation (as well as LT and $\dot{V}O_{2\max}$) in healthy young individuals.

Despite the strong coherence between individual SP and CP estimates, the power output – time integral above SP in the new ramp-sprint test was consistently less than W' (as derived from the traditional gold-standard method). Since the physiological determinants of W' are still not well understood (e.g. Ferguson et al. 2010; Skiba et al. 2014), the underlying reasons for this discrepancy are difficult to pinpoint. W' is mathematically equivalent to finite amount of work that can be performed above CP (Moritani

et al. 1981; Fukuba et al. 2003), and is generally thought to reflect the utilisation of a fixed anaerobic energy store and/or the accumulation of fatigue-related metabolites that cause a progressive reduction in work efficiency up to some 'critical' level (e.g. Poole et al. 1988; Coats et al. 2003; Fukuba et al. 2003; Jones et al. 2008; Murgatroyd and Wyld 2011; Skiba et al. 2014). Although there is thought to be no difference in W' between constant-power and ramp-incremental protocols, Morton et al. (1997) reported a wide within-participant variability of up to 10 kJ or 50 %. Similar discrepancies between W' estimated from the traditional and 3 min all-out tests have also been found, with relatively poor correlation between the measures [both negative ($r^2 = 0.51$) and positive ($r^2 = 0.03$ – 0.58) associations] and mean differences ranging approximately ± 30 % (Cheng et al. 2012; Dekerle et al. 2013; Fukuda et al. 2014). The precise meaning of these discrepancies is unclear. Using a single exercise test, an underestimation in W' would be expected to manifest as a corresponding overestimation in CP since the two parameters are co-dependent in the model, i.e. the end-exercise power output would be above CP were W' yet to be depleted (although this rarely appears to be the case; Cheng et al. 2012; Broxterman et al. 2013; Dekerle et al. 2013; Fukuda et al. 2014; present data). The W' estimate from either single or serial tests to establish the power-duration curve may be particularly sensitive to measurement error: with the test–retest coefficient of variation being 10–30 % (Vanderwalle et al. 1997; Johnson et al. 2011; Dekerle et al. 2013; de Lucas et al. 2013). Using the RST, we observed a narrow test–retest variation in W' estimation (<5 %), but with a difference between W' and work done above SP in the RST of ~ 20 % (3.7 kJ). Whether this relates to real differences in W' among constant-, variable- and incremental-power protocols (see Morton 2006), rather than accuracy of W' estimation per se, awaits resolution. Nevertheless, the apparently low W' estimate from the RST could be seen to represent a disadvantage of the approach over traditional methods of characterising the power-duration curve.

Application and optimisation of the RST protocol

The introduction of the 3 min all-out test (Burnley et al. 2006; Vanhatalo et al. 2007) represented a significant advance, since it appreciably reduced the testing burden for intensity domain assessment from typically five laboratory visits (one ramp-incremental and four constant-power tests) to as few as two (one ramp-incremental and one 3 min all-out test). Nevertheless the demands for the participant remain considerable with the 3 min all-out test, where failure to maintain $\dot{V}O_2$ above at least 95 % of the ramp-incremental determined $\dot{V}O_{2\text{peak}}$ for ~ 2.5 min deems the test invalid (Jones et al. 2010): this duration is required

to deplete W' , at least for healthy young participants (e.g. Vanhatalo et al. 2007).

With the RST protocol we hoped to build on the benefits of the 3 min all-out test by completing the two exercise phases (ramp and sprint) in immediate series, thereby necessitating only a single laboratory test for intensity domain assessment. But this is not the only potential advantage that was sought. The duration required for SP to stabilise during the sprint phase of the test was expected to be shorter than that during all-out exercise alone, because W' should have been depleted during the ramp-incremental phase of the RST (Morton et al. 1997). We found, on average, that SP could be established during the first minute of the RST sprint phase (from 30 to 60 s), and that there was no detectable change in SP thereafter (Figs. 1, 3). This provides the potential to reduce the duration for which maximal effort at close-to-maximal $\dot{V}O_2$ is required, and thereby reduce the burden of the testing for the participant.

Participants in the present study maintained close-to-maximal $\dot{V}O_2$ for ~3 min during the sprint phase of the RST. Although 7 of our 10 participants had not previously participated in a laboratory exercise test, all were habitually active and familiar with the sensations associated with exercise approaching $\dot{V}O_{2max}$. This may have contributed to their ability to endure the sustained discomfort of the sprint phase of the RST. A modified version of the RST may therefore benefit from a variable sprint phase duration, thereby reducing the testing burden and allowing the test to be more widely administered; at least in physically-active populations. SP could be determined after at least 30 s of maximal effort by the point at which SP stabilised and $\dot{V}O_2$ was maintained at its maximum. Such a modification would potentially retain the benefit of intensity domain demarcation and require only a short duration, e.g. 45–60 s, of sustained maximal effort at $\dot{V}O_{2max}$. Incorporating the flexibility to allow the SP duration to vary (after at least 30 s) would, we suggest, account for instances where SP did not rapidly stabilise—as occurred in one participant in our study. A shorter test would also provide the potential for more vulnerable populations such as children (c.f. Barker et al. 2012) and patients to be assessed using the RST approach. Whether this modification would impact the accuracy or precision of CP estimation remains to be determined, but it warrants further investigation.

Optimisation of the RST for delivery to patients, children, athletes or for use with different types of ergometry is not straightforward, however. In this test SP is determined by the product of a constant braking force on the ergometer flywheel and pedalling cadence (determined by the participant). A standardised procedure for setting the sprint phase ergometer resistance is required

under conditions where typically only participant physical characteristics are known. This is crucial because CP is sensitive to changes in pedal cadence during constant- and variable-power exercise (Hill et al. 1995; Barker et al. 2006; Vanhatalo et al. 2008b; Dekerle et al. 2013). The resistance must therefore be set a priori, but set such that the cadence at the (as yet unknown) CP will be close to the self-determined rate. We selected 80 rpm as a target cadence for SP determination and set the ergometer resistance based on the relationship observed between body mass (in kg) and CP in a cohort of 60 healthy active young men (van der Vaart et al. 2013) ($CP = 236 \pm 33$ W, range 168–321 W; body mass = 78 ± 9 kg, range 55–103 kg; $CP = 3.0 \pm 0.4$ W kg⁻¹, range 1.5–4.4 W kg⁻¹). Although this relationship was inherently variable ($r^2 = 0.32$), on average CP was approximated by three times body mass. The power value derived from this approximation was substituted in the equation to set the flywheel braking resistance at 80 rpm. Using this approach for the basis of setting the sprint phase resistance appears pragmatic because it resulted in cadences during the sprint phase of 81 ± 3 rpm (range 77–86 rpm). Applying the RST to patients, children or athletes would therefore first require the determination of a resistance factor calculation appropriate for each population. With these provisos, the RST test therefore offers promise for the assessment of exercise intensity domains in a single laboratory exercise test.

Conclusions

This study has demonstrated the efficacy of a novel ramp-sprint exercise testing protocol for estimation of LT, CP and $\dot{V}O_{2max}$ to within a tolerance of 5 % or less in a single laboratory test—a precision that compares favourably to current gold-standard methods. The RST was highly reproducible within participants and allowed accurate prescription of exercise intensity domains among individuals (Whipp 1996), which were verified in the responses of $\dot{V}O_2$ and blood [lactate] during independent constant-power tests. These data therefore suggest that the RST protocol allows non-invasive measurement of the exercise intensity domains in healthy young individuals in a single laboratory test, which may simplify the assessment, prescription and monitoring of exercise, exercise training, and therapeutic or experimental interventions.

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Conflict of interest The authors declare that they have no conflict of interest.

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